

Variational approximations of possibilistic inferential models¹²

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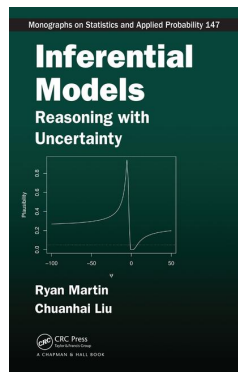
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¹Joint work with Leonardo Cella

²Extended version will be posted at [arXiv:2404.19224](https://arxiv.org/abs/2404.19224)

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- IMs offer reliable UQ for statistical inference/prediction
- I think IMs have advantages over other brands of UQ, e.g., Bayesian
- But Bayesian UQ has a long history, lots of computational developments
- Different computational tools needed?
 - IMs = possibilistic
 - Bayes = probabilistic
- My hope/hunch is that they're not entirely separate...



- Variational approximations are useful in Bayesian inference, offer simple solutions avoiding MCMC
- Here we lean on variational-like approximations too
- Basic idea goes as follows:
 - IM's credal set contains confidence dist'ns (CDs)
 - given α , consider a “most diffuse” $100(1 - \alpha)\%$ CD
 - approximate this CD by a parametric family, optimization
- Our idea is motivated in part by
 - Jiang, Liu & Zhang [arXiv:2408.16763](#)
 - Syring & M. [arXiv:1509.00922](#)

- Statistical model: $(Y \mid \Theta = \theta) \sim P_\theta$
- Goal is to infer Θ when $Y = y$ is given
- If L_y is the likelihood, then the relative likelihood is

$$R(y, \theta) = \frac{L_y(\theta)}{\max_{\vartheta} L_y(\vartheta)}, \quad \theta \in \mathbb{T}$$

- $\theta \mapsto R(\theta, y)$ defines a possibility contour, but inference based directly on this is lacking certain essential properties
- So, I apply the possibility transform to R :

$$\pi_y(\theta) = P_\theta\{R(Y, \theta) \leq R(y, \theta)\}, \quad \theta \in \mathbb{T}$$

Valid and efficient imprecise-probabilistic inference
with partial priors, II. General framework

Ryan Martin*

- IM is the map $y \mapsto (\underline{\Pi}_y, \overline{\Pi}_y)$ where
 - $\overline{\Pi}_y$ is the possibility measure determined by π_y
 - $\underline{\Pi}_y$ is the corresponding consonant belief function
- Lots of desirable properties,⁴ including *validity*

$$\sup_{\Theta \in H} P_{\Theta}\{\overline{\Pi}_Y(H) \leq \alpha\} \leq \alpha, \quad \alpha \in [0, 1], \quad H \subseteq \mathbb{T}$$

- Among other things, validity implies that IM-based tests and confidence sets have proper error rate control

⁴M. arXiv:2211.14567

- Downside is that computation of π_y can be a challenge
- Roughly, the best generally-applicable strategy we have is naive Monte Carlo:
 - for each θ on a suitably fine grid in $\mathbb{T}$
 - sample $\{Y_\theta^{(m)} : m = 1, \dots, M\}$ from P_θ
 - approximate the expectation by an average

$$\pi_y(\theta) \approx \frac{1}{M} \sum_{m=1}^M 1\{R(Y_\theta^{(m)}, \theta) \leq R(y, \theta)\}$$

- This is doable, but the “suitably fine grid” can be quite large in multi-parameter problems, expensive simulations
- Any other ideas?

- M. and Williams offer a simple Gaussian approx'n to the IM, but “large-sample” is arguably too idealistic
- Another way to get a parametric approximation?
- Our idea is inspired by
 - variational approx'ns common in Bayesian inference
 - recent work of Jiang, Liu, & Zhang (arXiv:2408.16763)
- High-level summary of our proposal:
 - choose a parametric family of distributions on $\mathbb{T}$
 - among those, find one that suitably aligns/agrees with π_y
 - approximate π_y by a prob-to-poss transform⁵

⁵Prob-to-poss transform might be available in closed-form...

- Recall the following fact:
 - if $\mathcal{C}(\overline{\Pi}_y)$ is the credal set corresponding to $\overline{\Pi}_y$
 - then it has a simple-to-state characterization

$$Q_y \in \mathcal{C}(\overline{\Pi}_y) \iff Q_y\{\theta : \pi_y(\theta) > \alpha\} \geq 1 - \alpha, \forall \alpha \in [0, 1]$$

- Best inner approximation,⁶ if it exists, is a Q_y^* such that “ $= 1 - \alpha$ ” is achieved for all α
- Easier: find $Q_y^{*\alpha}$ such that equality holds for a particular α
- Given such a $Q_y^{*\alpha}$, approximate contour as

$$\pi_y(\theta) \approx \underbrace{Q_y^{*\alpha}\{q_y^{*\alpha}(\Theta) \leq q_y^{*\alpha}(\theta)\}}_{\text{prob-to-poss transform of } Q_y^{*\alpha}}, \quad \theta \in \mathbb{T}$$

⁶For some models, Q_y^* is the Bayes/fiducial dist'n (M. *ISIPTA*'23)

Variational approximation, cont.

- Introduce a parametric family $\{Q_y^\omega : \omega \in \Omega\}$
- Goal: find ω^* that solves the equation

$$\underbrace{Q_y^\omega \{\theta : \pi_y(\theta) > \alpha\} - (1 - \alpha)}_{f_\alpha(\omega)} = 0$$

- Monte Carlo approximation:

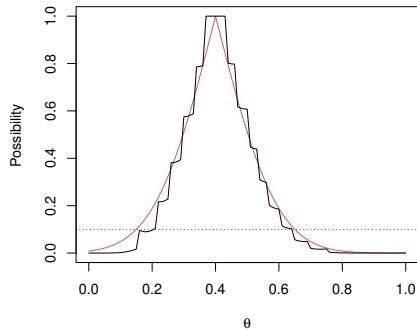
$$\hat{f}_\alpha(\omega) = \frac{1}{M} \sum_{m=1}^M 1\{\pi_y(\Theta_\omega^{(m)}) > \alpha\} - (1 - \alpha), \quad \Theta_\omega^{(m)} \stackrel{\text{iid}}{\sim} Q_y^\omega$$

- *Stochastic approximation* to solve the equation $\hat{f}_\alpha(\cdot) = 0$

- Summary:
 - choose a parametric family indexed by ω
 - find ω to achieve credal condition at given α
 - approximate contour by prob-to-poss transform
- Taking Q_y^ω as Gaussian makes sense (cf. M. & Williams), e.g., $Q_y^\omega = N(\hat{\theta}_y, \omega \hat{J}_y^{-1})$, but not the only option
- New approach still requires Monte Carlo evaluation of π_y
 - in our experience, fewer evaluations required
 - much more efficient than naive approach above

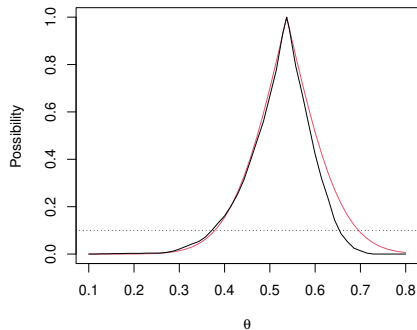
Example: binomial probability

- $(Y \mid \Theta = \theta) \sim \text{Bin}(n, \theta)$
- $n = 15$ and $y = 6$
- Gaussian approximation, $\alpha = 0.1$



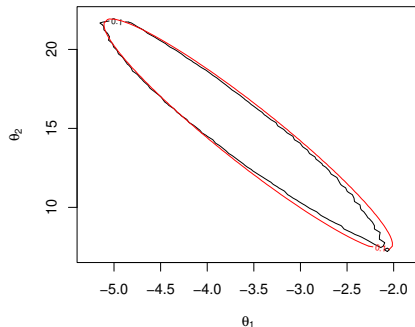
Example: bivariate normal correlation

- Y^n are iid bivariate normal, mean 0, var 1, cor Θ
- $n = 50$, $\Theta = 0.5$
- Gaussian approximation, $\alpha = 0.1$



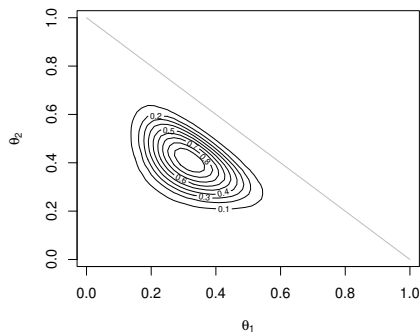
Example: logistic regression

- Real-data illustration: one feature, $n = 120$ samples
- $(Y_i \mid X_i = x_i, \Theta = \theta) \stackrel{\text{ind}}{\sim} \text{Ber}\{\text{logit}^{-1}(\theta_0 + \theta_1 x_i)\}$
- Gaussian approximation, $\alpha = 0.1$



Example: multinomial prob vector

- $Y \sim \text{Mult}_K(n, \Theta)$
- $K = 3$, $n = 25$, $Y = (8, 10, 7)$
- Dirichlet approximation, $\alpha = 0.1$



- Efficient IM computation is a challenge
- Naive Monte Carlo works in principle, just expensive
- Our basic idea:
 - find a simple probability dist'n to represent IM's credal set
 - approximate IM's contour by the prob-to-poss transform of the representative
- Things to do (in the extended version):
 - more multi-parameter cases
 - marginal inference
- Key open questions:
 - can we stitch together approximations for different α ?
 - any better ideas?



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

`https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/`

- Papers, talk slides, etc. can be found on my website:

`www4.stat.ncsu.edu/~rmartin/`

- Email me (`rgmarti3@ncsu.edu`) if you have questions, etc

Thanks for your attention!